TROPIC Controlled Clinical Analysis Report

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Study EFC6193 / XRP6258 -- Abbreviated Clinical Study Summary

Current controlled release: tag v0.3.0-clinical-simulation docs/RELEASE_NOTE_v0.3.0-clinical-simulation.md controlled clinical-submission simulation

SAP v4.0 lock note (2026-06-25): This report is a generated demonstration output under remediation control. It is not a clinical study report for submission and it is not a source of analysis requirements. The governing analysis plan is 02_specifications/sap/TROPIC_SAP_v4.0_industry_grade.docx.

Study Title: Prednisone plus Cabazitaxel or Mitoxantrone for mCRPC Progressing After Docetaxel

Sponsor: Sanofi-Aventis | Phase: III, Open-Label RCT | NCT: 00417079

Publication: de Bono JS et al., Lancet 2010;376(9747):1147-1154

Mixed real + synthetic data. All MP-arm statistics (N=371) are derived from the official Sanofi de-identified SDTM public data release (2013) -- real patient-level data. The Cabazitaxel (CbzP, N=378) arm is a synthetic, illustrative cohort built by two methods depending on endpoint: OS and PFS are reconstructed via genuine Guyot (2012) IPD reconstruction from the published Lancet 2010 Kaplan-Meier curves (independent of the MP arm, so the reconstruction is non-circular); the secondary time-to-event endpoints (TTPSA/TTUMOR/TTPAIN) are PH-scaled from the MP arm and are circular by construction; the non-TTE domains are fixed-seed sampled from published Lancet 2010 Table 1/Table 2 marginals. The live mixed-source OS comparison remains compatible with the publication, but the corrected real-MP PFS derivation yields HR 0.87 and does not reproduce the published PFS effect. The comparator is not real patient data and is shown only to exercise the analysis pipeline, not as a clinical finding (see ADRG Section 7).

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1. Study Population

The Safety Population consisted of 371 patients in the Mitoxantrone + Prednisone (MP) arm, all with metastatic castration-resistant prostate cancer (mCRPC) progressing during or after docetaxel-based chemotherapy.

Characteristic	MP Arm (N=371)
ECOG Performance Status >=1	Majority
Prior docetaxel progression	100% (eligibility criterion)
Measurable disease	Subset (MEASDISF = Y)
Visceral disease	Subset (VISCFL = Y)
Baseline PSA (median)	~110 ng/mL
Baseline ALP (median)	~140 U/L

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2. Overall Survival & Progression-Free Survival -- Guyot Reconstruction (synthetic comparator)

The primary survival endpoints exercise the stratified Cox / log-rank machinery and the hierarchical step-down gatekeeping logic (ICH E9). For OS and PFS the synthetic CbzP arm is recovered by genuine Guyot (2012) IPD reconstruction (IPDfromKM) from the published de Bono 2010 Kaplan-Meier curves (Fig 2A = OS, Fig 3 = PFS) plus the transcribed numbers-at-risk tables. The survival shape comes from the published curve itself -- independently of the MP arm (no hazard-ratio scaling), so the CbzP-vs-MP hazard ratio is not circular: it emerges from the reconstructed CbzP curve versus the real MP data.

Statistic	**Synthetic CbzP (N=378, Guyot)(1)**	**Real MP (N=371)**	**Published (de Bono 2010)**
Median OS	15.2 months	12.7 months	15.1 mo vs 12.7 mo
Median PFS	2.7 months	1.4 months	2.8 mo vs 1.4 mo
OS HR (CbzP vs MP)	0.71 (95% CI 0.60-0.85)	Reference	0.70 (0.59-0.83)
PFS HR (CbzP vs MP)	0.87 (95% CI 0.75-1.02)	Reference	0.74 (0.64-0.86)

OS/PFS are reconstructed, not PH-scaled. The intrinsic curve-reconstruction gates pass. Using the live stratified TFL method, OS is compatible with the publication (HR 0.71), while PFS is outside the legacy compatibility range (HR 0.87; range 0.64-0.84). The current real-MP PFS derivation is intentionally restricted to typed RECIST/PSA/F-042 pain/death components and excludes exploratory bone/clinical-progression signals. This mixed-source PFS comparison is therefore a disclosed pipeline diagnostic, not evidence that the published effect was reproduced; see 01_source_data/guyot_validation_report.md.

(1) Synthetic, illustrative -- not real patient data; reconstructed from the published KM curves (Guyot 2012).

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3. Secondary Time-to-Event & Response Endpoints -- Pipeline Demonstration (synthetic comparator)

The secondary time-to-event endpoints (TTPSA, TTUMOR, TTPAIN) use the synthetic CbzP arm PH-scaled from the real MP arm -- no published KM curves with at-risk tables exist for these, so Guyot reconstruction is not possible and their HRs/p-values are circular by construction (non-inferential). Response endpoints use the fixed-seed simulated CbzP cohort. Values are the live output of 05_outputs/tfl/tfl_generation.R (see 05_outputs/tfl/output/tables/T-11-Efficacy_Tables.txt).

Endpoint	**Synthetic CbzP(1)**	**Real MP**	**Pipeline HR / test**	**Pipeline p**	**Published (de Bono 2010)**
Time to PSA Progression(2)	2.7 mo (286/378)	2.2 mo (265/371)	stratified Cox HR 0.84 (0.70-0.99)	0.0319	median 6.4 mo, HR 0.75

Endpoint	Synthetic CbzP(1)	Real MP	Pipeline HR / test	Pipeline p	Published (de Bono 2010)
Time to Tumour Progression(2)	4.9 mo (166/378)	5.0 mo (96/371)	stratified Cox HR 0.89 (0.66-1.22)	0.4406	publication-era endpoint context
Time to Pain Progression(2)	8.1 mo (130/378)	NE (37/371)	stratified Cox HR 2.85 (1.98-4.12)	<0.0001	SAP T-11-8 mapping restored
PSA Response (>=50% decrease; baseline PSA >=20 ug/L)	40.2% (145/361)	18.5% (61/329)	Fisher's exact	5.2e-10	39% vs 24% (p = 0.0002)
Overall Response Rate (ORR, confirmed)Section	16.8% (30/179)	5.9% (12/203)	Fisher's exact	0.0009	14.4% vs 4.4% (p = 0.005)
Pain Response (F-042)	N/A (PN unavailable)	27.6% (43/156)	descriptive responder rate	--	source-qualified PN/SV implementation

(1) Synthetic, illustrative -- not real patient data. (2) PH-scaled from the real MP arm where applicable; any HR is circular by construction (descriptive of synthetic data only, not a treatment-effect estimate). Section ORR is restricted to the measurable-disease subpopulation (CbzP N=179, MP N=203). TTUMOR is ITT-primary (CbzP N=378, MP N=371), with measurable disease retained as supportive. Pain response is unavailable for CbzP because no PN domain is reconstructed. The OS/PFS primary endpoints -- which are non-circular Guyot reconstructions -- are in Section 2.

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4. Safety -- Adverse Events (Safety Population)

Treatment-emergent adverse events (TEAEs) were defined as events occurring on or after the first dose date (TRTEMFL = "Y" in ADAE).

4.1 Overall TEAE Incidence

Category	Synthetic CbzP (N=371)	%	Real MP (N=371)	%
Any TEAE	364	98%	328	88%
Any Grade >=3 TEAE	310	84%	147	40%
Any Serious TEAE (SAE)	145	39%	78	21%
Any TEAE leading to discontinuation	68	18%	32	9%

4.2 Grade >=3 TEAEs by System Organ Class (Top 6)

System Organ Class	CbzP (n, %)	MP (n, %)
Blood & Lymphatic System Disorders	293 (79%)	39 (11%)
Gastrointestinal Disorders	34 (9%)	6 (2%)
General Disorders & Admin Site Conditions	35 (9%)	36 (10%)
Musculoskeletal & Connective Tissue Disorders	18 (5%)	35 (9%)
Infections & Infestations	8 (2%)	19 (5%)
Nervous System Disorders	4 (1%)	14 (4%)

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5. Laboratory Toxicity -- CTCAE Grade Shift

Baseline to worst post-baseline toxicity-grade shifts use the one-sided ADaM baseline flag ABLFL, the analysis record flag ANL01FL, and ATOXGR in ADLB. For the real MP arm, ATOXGR carries the source LBTOXGR; the analysis pipeline does not independently re-grade laboratory values against CTCAE thresholds. The CbzP laboratory arm remains synthetic and illustrative.

5.1 ANC / Neutrophils -- Key Finding

In the synthetic CbzP laboratory cohort, 321/371 (86.5%) had a worst post-baseline Grade 3/4 neutrophil result, compared with 125/371 (33.7%) in the real MP arm.

5.2 Haemoglobin

Worst post-baseline Grade 3/4 haemoglobin results occurred in 34/371 (9.2%) synthetic CbzP subjects and 7/371 (1.9%) real MP subjects.

5.3 Platelets

Worst post-baseline Grade 3/4 platelet results occurred in 16/371 (4.3%) synthetic CbzP subjects and 3/371 (0.8%) real MP subjects.

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6. Exposure Analysis (ADEX)

Cycle-by-cycle exposure was captured in ADEX across parameters including:

- RDI (Relative Dose Intensity) -- source EXTRINT carried at subject level only when repeated values are internally consistent; no dose-intensity formula is reconstructed
- NCYCLE -- count of distinct cycles with a positive administered qualifying IV dose
- CUMDOSE -- source cumulative administered IV dose (EXCUMD2)
- NDELDOSE / NREDDOSE -- counts of source-indicated delays and qualifying dose reductions

FDA Project Optimus alignment: the subject-level all-cycles RDI is paired with the Cycle 1 ANC nadir in the exploratory E-R scatter plot (F-17-1). This is a non-confirmatory pipeline demonstration, not a dose-optimization conclusion.

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7. Pipeline Technical Summary

Layer	Technology	Standard
Source Data	SAS7BDAT (Sanofi / Project Data Sphere)	CDISC SDTMIG v3.1.1 (trial-era)
ADaM Production	SAS 9.4	ADaMIG v1.3
Independent Validation	R 4.6.0 / Pharmaverse	ADaMIG v1.3
Reconciliation	diffdf package	100% cell-by-cell match
TFL Generation	ggplot2, survival, patchwork	ICH E3 / NEJM style
Orchestration	Python 3.10+ (cibuild.py)	Manifest-driven 40-stage CI pipeline
Simulation methods annex	Python 3.12.13, NumPy 2.2.6, float64/PCG64	Frozen data-free MAP, 400,000 replicates, independent evidence verification

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8. Informative Simulation Methods Annex

The submission-style package includes a separately governed, data-free fixed-design time-to-event methods evaluation. Its Model Analysis Plan and Model Analysis Report are generated from a frozen YAML protocol and authoritative aggregate JSON. Across ten scenarios, 400,000/400,000 requested replicates completed with no failures. The two analytic-null estimates were 2.525% and 2.504% at one-sided alpha 2.5%, with MCSE 0.000496 and 0.000494; both prespecified null checks passed.

This annex is deliberately separate from the clinical analyses above. It uses a public 377/378 design calibration and engineering stress assumptions, not authoritative TROPIC IPD; it omits original-trial stratification and has no sponsor-approved minimum effect or power threshold. Its evidence qualification remains NOT_QUALIFIED: informative, non-MIDD, non-confirmatory, and unsuitable for clinical or filing decisions.

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9. Data Provenance & Limitations

Real data (MP arm): All 371 MP-arm patients, 5,428 AE records, 266 OS events, and ~79,000 laboratory records are derived directly from the official Sanofi de-identified public SDTM release (dated June 2013).

Synthetic comparator (CbzP arm): The Cabazitaxel arm was not included in the Sanofi public data release. The CbzP arm used in figures and comparative tables is synthetic and illustrative, built by two methods depending on endpoint:

- OS and PFS are reconstructed via genuine Guyot (2012) IPD reconstruction (IPDfromKM) from the published de Bono 2010 KM curves (Fig 2A = OS, Fig 3 = PFS) + transcribed numbers-at-risk tables. The shape comes from the published curve independently of the MP arm, so the reconstruction is non-circular. Intrinsic curve gates pass; the live stratified OS diagnostic is compatible (HR 0.71), while the mixed-source PFS diagnostic is outside its legacy range (HR 0.87) and is explicitly a warning in 01_source_data/guyot_validation_report.md.

- Secondary TTE endpoints (TTPSA, TTUMOR, TTPAIN) remain PH-scaled from the real MP arm (no published KM curves exist for them) and are circular by construction.

- Non-TTE domains (AE, laboratory, exposure, demographics) are fixed-seed sampled from published Lancet 2010 Table 1/Table 2 marginal distributions.

The arm is not real patient data; secondary CbzP-vs-MP comparisons are illustrative only. See ADRG Section 7 for the full reconstruction methodology.

Single source of truth. Every count, percentage, median, HR and p-value in this report is produced by 05_outputs/tfl/tfl_generation.R and written to 05_outputs/tfl/output/tables/.txt. Narrative numbers are transcribed from those files; the generated tables govern in case of any discrepancy.

Reference

de Bono JS, Oudard S, Ozguroglu M, et al. Prednisone plus cabazitaxel or mitoxantrone for metastatic castration-resistant prostate cancer progressing after docetaxel treatment: a randomised open-label trial. Lancet. 2010;376(9747):1147-1154.

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